

DO POWER SAVERS REALLY SAVE POWER?

Many companies have found market for their small handheld products that claim to save 30-40 per cent electrical power, thereby reducing electricity bill. But can you really save electricity by plugging in a small device?



A power saver unit plugged into a wall socket

■ SHWETA DHADIWAL BAID

It is rightly said, "Energy saved is energy produced." Power saver is a smart attempt in this direction.

The theory behind power savers or energy savers (used interchangeably by many people) is foolproof, but there are a lot of conditions that affect the practical applications of power saver

devices. Here's what we asked the manufacturers...

What is a power saver?

A power saver (PS) device resembles a mosquito-repellent plugin gadget (like All-out or GoodKnight) and is meant to save power in houses, shops and small offices. It is plugged into a socket to work in parallel with other appliances connected to the electrical

circuit of an establishment.

What are the benefits of power savers?

Power savers primarily save electrical power. However, these also improve the quality of input electrical signal, and thus enhance the life-span of household appliances.

How do they work?

The design of a power saver is based on the principles of power factor correction (PFC), harmonic current filter and surge protection. While PFC helps in reducing reactive power wastage, harmonic reduction improves the quality of power signal.

In the long run, surges may cause heating and seriously damage the household devices. The improved quality of the signal prevents damage to the transmission line. The circuit of a power saver is designed to prevent heat loss by appliances.

What is inside a power saver?

The device has a big PFC capacitor and a current harmonic filter circuit. Some devices also have metal-oxide varistors (MOVs) that act as transient suppressors, protecting the circuit from voltage spikes. A series connected thermal fuse is used with MOVs to provide protection in the event of short circuit. Power savers come with integrated voltage stabilisation circuit too. A light-emitting diode (LED) is used to indicate power-on condition.

Is it legal to use power savers?

Yes, it is absolutely legal. Using a power saver does not mean fiddling